Krakken-2048: Experimental Verification and Distinguisher Battery

Abstract

This note is a companion to the active-S-box bound analysis of Krakken-2048. It records two categories of result obtained by direct computation against the reference implementation: (i) *exact proofs* of the cryptographic foundations the bound analysis relies on (the MDS branch number, the S-box's differential and linear properties, and the bijectivity of the word-permutation layers), each established by complete enumeration; and (ii) an *empirical distinguisher battery* (avalanche, rotational and RX-difference, integral/saturation, cube / algebraic degree, one- and two-sided zero-sum, exact algebraic-degree growth, differential clustering, impossible differentials, boomerang / BCT, differential-linear, and invariant-subspace / S-box-coset structure) run against the compiled permutation. The foundations are proven. The distinguisher battery finds no distinguisher surviving the full eight rounds in any tested family. We are explicit throughout that the battery is empirical and necessary-but-not-sufficient: each test rules out a *detectable* weakness of its kind at the sampled scale, not the existence of a more subtle attack, and several have stronger (MILP-based) forms that remain future work.

1 Exact proofs of the foundations

The following properties underlie every active-S-box bound in the companion analysis. They had previously been inherited by analogy from AES/Whirlpool; here each is verified exhaustively for the constants as implemented.

Property	**Method**	**Result**
MDS branch number	all 12,869 square submatrices over $GF(2^8)$	9 (maximal)
S-box bijection	full table check	permutation of $0-255$
S-box diff. uniformity	full 256×256 DDT	$4 \Rightarrow \mathrm{DP} = 2^{-6}$
S-box linearity	full 256×256 LAT	bias $16 \Rightarrow \mathrm{corr}^2 = 2^{-6}$
S-box boomerang uniformity	full 256×256 BCT	$6 \Rightarrow$ switch $2^{-5.42}$
π word permutation	index-map bijection check	bijection
InkCloud word permutation	index-map bijection check	bijection
InkCloud multiplier	$\gcd(7, 32) = 1$	invertible, $7^{-1} = 23 \bmod 32$

Table 1: Exhaustively-proven foundational properties (reduction polynomial `0x11D`; MDS row $(1, 1, 4, 1, 8, 5, 2, 9)$; ABYSSAL S-box).

These establish that the per-S-box weight is exactly 2^{-6} for both differential and linear cryptanalysis, that the diffusion matrix achieves the maximal branch number 9, and that the permutation layers preserve the full state (are bijective). The inverse permutation was additionally implemented and verified to round-trip bit-exactly with the forward permutation in both directions at all eight round counts.

2 Empirical distinguisher battery

Each test below was run against the compiled reference permutation. For each we state what a positive (weak) result would look like and what was observed.

2.1 Avalanche (strict diffusion)

Flipping a single input bit should change about half of the 2048 output bits. Measured over 200 trials per round:

Round	mean	min	max	std
1	0.4996	0.4673	0.5332	0.0114
2	0.4998	0.4707	0.5454	0.0119
3	0.4990	0.4639	0.5283	0.0107
4	0.5004	0.4634	0.5322	0.0115
5	0.5009	0.4604	0.5283	0.0112
6	0.5004	0.4722	0.5386	0.0108
7	0.4994	0.4712	0.5312	0.0113
8	0.4997	0.4717	0.5337	0.0119

Table 2: Fraction of output bits changed by a one-bit input flip. Full avalanche (mean within 1% of 0.5) is reached at round 1 and held through round 8.

2.2 Strict avalanche criterion, completeness, and monobit balance

The avalanche test above reports only the *mean* fraction of output bits flipped. A mean of exactly 0.5 is, however, compatible with a badly structured per-bit dependency: an input bit that always flips a given output bit, and never flips another, averages to 0.5 while violating diffusion. To see this we measure the full strict-avalanche-criterion (SAC) matrix (Webster and Tavares, 1986): for every pair (input bit i, output bit j),

$$\mathrm{SAC}[i][j] \;=\; \Pr\big[\text{output bit } j \text{ flips} \;\big|\; \text{input bit } i \text{ is flipped}\big] \qquad (\text{ideal } \tfrac{1}{2}),$$

estimated over $N = 4096$ random base states per input bit ($2048 \times 2048 = 4.19 \times 10^6$ cells, $2 \times 2048 \times N$ permutation calls per round). From the same data we extract two derived properties: *completeness* (every cell strictly between 0 and N, i.e. every input bit influences every output bit but none deterministically) and a per-output-bit *monobit* frequency over random inputs (ideal $\frac{1}{2}$). At $N = 4096$ the two-sided 5σ threshold on a single proportion is $|p - \frac{1}{2}| > 0.0391$.

The SAC matrix is statistically indistinguishable from ideal at *every* round including round 1: the mean cell deviation is exactly the random-sampling value, and the one-to-five cells exceeding 5σ are consistent with the ≈ 2.4 false positives expected over 4.19×10^6 cells. The completeness count is decisive: with zero dead and zero stuck cells at all rounds, every one of the 2048 input bits provably influences every one of the 2048 output bits, and none does so deterministically—the per-bit form of the full-state first-round diffusion claimed by the mean-avalanche test. No output-bit monobit bias was detected at any round. This is a stronger negative result than the mean avalanche: a permutation with a correct mean but a structured dependency matrix would fail here, and none of the 2048×2048 couplings deviates beyond sampling noise.

Round	SAC max $\lvert p-\frac{1}{2}\rvert$	SAC mean $\lvert p-\frac{1}{2}\rvert$	cells $> 5\sigma$ (exp. ≈ 2.4)	dead / stuck cells	monobit max $\lvert f-\frac{1}{2}\rvert$
1	0.0415	0.0062	4	0 / 0	0.0276
2	0.0405	0.0062	2	0 / 0	0.0295
3	0.0422	0.0062	5	0 / 0	0.0305
4	0.0422	0.0062	2	0 / 0	0.0249
5	0.0413	0.0062	3	0 / 0	0.0264
6	0.0427	0.0062	1	0 / 0	0.0293
7	0.0435	0.0062	4	0 / 0	0.0271
8	0.0413	0.0062	4	0 / 0	0.0303

Table 3: SAC matrix, completeness, and monobit balance ($N = 4096$ per input bit, $5\sigma = 0.0391$). "SAC mean $\lvert p-\frac{1}{2}\rvert$" is the mean absolute deviation across all 4.19×10^6 cells; the value 0.0062 is exactly that expected of an ideal Binomial($N, \frac{1}{2}$) proportion ($\approx 0.399\,\sigma$). "cells $> 5\sigma$" tracks the random-expected 2.4 false positives. "dead" cells never flip (incomplete diffusion), "stuck" cells always flip (deterministic coupling); both are zero at every round.

2.3 NIST SP 800-22 statistical test suite on the sponge keystream

The probes above target the bare permutation. We additionally subjected the full sponge *keystream*—the permutation as used in the stream-cipher mode of the Krakken-Disk volume encryptor—to the NIST Statistical Test Suite (SP 800-22). Three independent 5500 MB encrypted volume containers were produced. Each container is initialised *empty*, so its plaintext is essentially the all-zero sector image; the encryptor forms $ciphertext_i = plaintext_i \oplus \mathrm{Sponge}(key \,\|\, nonce \,\|\, \mathrm{LE64}(i))$, so up to the per-sector random nonces (24 bytes per 4096-byte sector) and a small fixed header the ciphertext *is* the Krakken sponge keystream. Testing the ciphertext of an empty container is therefore, in effect, testing the raw keystream, with no plaintext entropy masking the permutation.

Each file was assessed over 10 sequences of 10^6 bits (the standard STS configuration). All fifteen test families—Frequency, Block Frequency, Cumulative Sums, Runs, Longest Run, Binary Matrix Rank, Spectral (FFT), Non-Overlapping Template (148 templates), Overlapping Template, Maurer's Universal, Approximate Entropy, Random Excursions and its Variant, Serial, and Linear Complexity—passed for all three files: every test's proportion of passing sequences met or exceeded the NIST minimum (8/10, or 5/6 for the random-excursion tests), and every uniformity P-value was non-degenerate.[1]

As an independent high-throughput cross-check, a custom monobit and runs test was run over a larger 10^8-bit window of each file (Table 4); all P-values are unremarkable and every byte-level Shannon entropy exceeded 7.999 bits/byte.

This is a mode-level empirical randomness check on the sponge keystream: like the rest of the battery it is necessary but not sufficient, ruling out detectable first- and second-order statistical bias at the sampled scale ($\approx 10^7$–10^8 bits) rather than establishing a security property of any construction built on the permutation.

[1] The suite overwrites its summary report on each run; the complete fifteen-family report is retained for `random3.bin` (no failures) and the other two files were logged at the same all-pass level.

File	Monobit P	Runs P	Entropy (bits/byte)
`random1.bin`	0.7165	0.5258	> 7.999
`random2.bin`	0.2247	0.3089	> 7.999
`random3.bin`	0.9925	0.5349	> 7.999

Table 4: High-throughput monobit and runs cross-check over 10^8 bits per file. All P-values exceed the 0.01 significance threshold.

2.4 Sponge-hash collisions and output distribution

The NIST suite above tests the keystream; here we test the sponge *hash* (256-bit digest, rate 160 bytes) for collision and distribution behaviour, a family the rest of this note does not cover. Three quick probes were run against the compiled hash.

Birthday collisions. We hashed $N = 10^6$ random messages, truncated each digest to t bits, and counted colliding pairs. For an ideal random function the expected number of colliding pairs is $\binom{N}{2}/2^t \approx N^2/2^{t+1}$; a structural collision weakness would appear as a large excess at some truncation width.

t (bits)	expected	observed	obs/exp
28	1862.6	1835	0.985
30	465.7	495	1.063
32	116.4	122	1.048
34	29.1	27	0.928
36	7.28	5	0.687
38	1.82	2	1.100
40	0.46	1	2.199

Table 5: Truncated-digest birthday collisions, $N = 10^6$ messages. Observed counts track the ideal $N^2/2^{t+1}$ closely wherever the expectation is statistically meaningful ($t \leq 34$); the apparent ratios at $t \geq 36$ are small-count Poisson fluctuations (expectation below 7). No truncation width shows a collision excess.

Output diffusion / near-collisions. For 2×10^5 random message pairs differing in a single input bit, the full 256-bit digest difference had mean Hamming weight 127.98 (ideal 128) and standard deviation 8.00 (ideal $\sqrt{256 \cdot \frac{1}{4}} = 8$); the minimum over all pairs was 93 and the maximum 165. The minimum staying far from 0 rules out cheap one-bit near-collisions.

Digest byte uniformity. Over 2×10^5 digests (6.4×10^6 output bytes), the byte-value frequencies gave $\chi^2 = 273.8$ on 255 degrees of freedom (ideal 255 ± 22.6; $z = +0.83$), consistent with a uniform output distribution.

All three probes are consistent with an ideal random function; as elsewhere they are empirical and sampling-bounded, excluding gross collision or distribution defects at the $\sim 10^6$-message scale rather than establishing collision resistance analytically.

2.5 Rotational

Rotational cryptanalysis exploits that XOR, AND, and bit-rotation commute with rotating the whole state. We compare rotate($P(X)$) against P(rotate(X)): a rotation-invariant design makes these equal, a sound design makes them differ in ~ 1024 of 2048 bits (random). Measured mean Hamming distance was 1022.9–1025.2 across rounds 1–8 for rotation amount 1, and 1022.9–1024.9 at eight rounds across rotation amounts $\{1, 7, 13, 17, 32, 63\}$. No rotational relation was detected at any round; the relation is destroyed from round 1, consistent with the non-symmetric round constants and the carry behaviour of the PRESSURE additions breaking rotational invariance.

2.6 Integral / saturation

Saturating one input byte through all 256 values and XOR-summing the outputs should yield a balanced (zero-sum) signature in many output bytes if the design has a surviving integral property. Random expectation is ~ 1 zero byte of 256. Measured averages were 0.25–1.25 zero bytes across rounds 1–8 (samples at or below the random baseline at every round). No integral distinguisher was found at any round—including round 1, a stronger outcome than AES, whose classical integral survives three rounds; Krakken's full-state first-round diffusion removes the balanced property immediately.

2.7 Cube / algebraic degree

XOR-summing the output over all assignments of a k-bit input cube reveals the top-degree ANF coefficient; a structural zero across many constant settings is a cube distinguisher. Using $k = 8$ and requiring the cube-sum bit to vanish across 20 random constants (chance baseline $2048 \cdot 2^{-20} \approx 0.002$, so any survivor is real signal), the count of consistent-zero output bits was:

Round	consistent-zero bits (of 2048)
1	32
2	0
3	0
4	0
8	0

Table 6: Cube / algebraic-degree distinguisher. A genuine distinguisher shows a count that decays to zero as algebraic degree builds; here it does so by round 2. The round-1 value (32) is a real but expected low-degree artifact of a single round, far below the eight-round permutation.

2.8 Zero-sum (forward and two-sided)

A zero-sum set has both input-XOR and output-XOR equal to zero. The *forward* test built affine input subspaces (input-XOR $= 0$ by construction, dimension 8, 24 subspaces) and found no round at which the output-XOR was also zero (0 full zero-sums, 0 consistent-zero bits at every round). The *two-sided* test (using the verified inverse permutation, dimension 9, 12 subspaces) placed a subspace at every split $r_{\text{back}} + r_{\text{fwd}} = 8$ and required input-XOR and output-XOR to vanish simultaneously. Only the trivial edge splits ($0+8$ and $8+0$, in which one side is the raw subspace) showed any zero-sum; the degree saturates within two rounds from each side, and no split with $r_{\text{back}} \geq 1$ and $r_{\text{fwd}} \geq 1$ produced a two-sided zero-sum. No zero-sum distinguisher was found over the full eight rounds.

2.9 Exact algebraic degree

This is the test most capable of revealing a higher-order differential or cube weakness: rather than checking a fixed cube, it computes the *exact* algebraic-normal-form degree of each output bit as a Boolean function of a chosen set of $n = 10$ input variables (via the Möbius transform over all 2^{10} assignments, other inputs fixed). A degree that plateaus below the maximum n as rounds increase would be a distinguisher; a healthy permutation saturates to near-maximal degree within a few rounds. We tested six independent variable placements and report the weakest (minimum) output-bit degree, since an attacker chooses the variables that minimise it.

Round	**max degree**	**min degree (weakest bit)**	
1	10	5	degree still building
2	10	8	saturated
3	10	8	saturated
4	10	8	saturated
8	10	8	saturated

Table 7: Exact algebraic degree over $n = 10$ input variables (maximum possible degree 10). Degree reaches near-maximal by round 2 and holds.

The minimum degree of 8 from round 2 onward warranted scrutiny, since a plateau below maximal is the signature of a distinguisher. Direct inspection of the full degree distribution at eight rounds resolves it as benign: of the 2048 output bits, 1027 have degree 10, 1020 have degree 9, and exactly *one* has degree 8 (and that bit was verified to depend on all 10 variables, so its degree is not capped by missing dependence). This is precisely the distribution expected of 2048 near-random degree-≤ 10 Boolean functions: the top-degree ANF coefficient is present or absent with probability $\approx 1/2$, so a small number of bits randomly fall to degree 9 and, rarely, 8. A genuine weakness would instead show many output bits stuck at low degree with the count persisting or growing across rounds; the observed behaviour is the opposite (full saturation by round 2 with a single statistical straggler). No low-degree distinguisher is present over the tested variable sets.

2.10 Division-property integral (bit-based, MILP)

The integral test above (byte saturation) is a sampling test; its analytical counterpart is the *bit-based division property* (two-subset / conventional CBDP, Todo–Xiang et al.), which propagates a division trail through the round function as a mixed-integer program and proves an output bit *balanced* (zero-sum over the chosen input cube) exactly when its division trail is infeasible. Unlike the empirical battery, a bit certified balanced this way is a *proof*, not a statistical observation. This is the analysis previously listed as future work; we report it here as a working, sound harness together with its current results.

Model. One Krakken round is modelled bit-for-bit: the three GF(2)-linear layers (the MDS-based diffusion, the butterfly diffusion, and the InkCloud word-shuffle) by their exact matrices extracted from the reference implementation, the ARX Pressure layer by a sound full-adder carry-chain over-approximation, and each ABYSSAL S-box by its division-property transitions. The circuits are validated bit-for-bit against the compiled permutation and the ARX division rule against exhaustive enumeration of a 4-bit adder, so the model is *sound*: every bit it certifies balanced is a genuine zero-sum bit, and the conservative directions (the ARX over-approximation and the inequality S-box model below) can only cause it to *miss* balanced bits, never to invent one.

Making it tractable. The textbook "selector" S-box model encodes the ≈ 2015 minimal division trails of the 8-bit S-box as that many binary selector variables *per S-box instance*—about 5×10^5 binaries per round across the 256 S-boxes of the nonlinear layer. At that size the solver (SCIP) could not resolve a *single* output bit, even without a time limit (the one-round model has 937,952 variables; every output bit hit the per-bit time cap). We replace it with a compact *inequality* model: a set of linear inequalities over the 16 input/output bits with *no* auxiliary variables, generated as one "no-good" exclusion cut per impossible division transition and then greedily merged. The result is 661 inequalities and is *exact*—verified over all 2^{16} points of the S-box division polytope to retain every one of the 62,741 valid transitions and exclude all 2,795 impossible ones, reproducing the trail table and the algebraic degree 7. This cuts the one-round model from 937,952 to 422,112 variables and is what makes the program solvable at all.

Current result. With the compact model the round-1 program solves to optimality—for a saturated input word as the cube, output bit 0 is proved *reachable* (i.e. not balanced) in $\approx 160\,\mathrm{s}$, whereas the selector model never terminated on any bit. This is consistent with the empirical integral finding above: full-state first-round diffusion already removes byte balance, so no surviving integral is expected even at one round. The exhaustive determination—sweeping all 2048 output bits and increasing the round count until no balanced bit remains, which would fix the integral-distinguisher length—is a long-running computation; the operational harness and its round-1 confirmation are reported here, with the multi-round sweep to follow. The harness is released alongside this note so the result can be reproduced and extended.

2.11 Differential clustering (empirical differential probability)

The active-S-box bound in the companion analysis bounds the best single *characteristic*; a differential attack instead exploits a *differential* ($\delta_{\text{in}} \to \delta_{\text{out}}$), summing the probabilities of all characteristics sharing that input/output pair (the cluster). A differential can therefore have higher probability than any single characteristic. We probe this empirically: fix the minimal-trail input difference identified in the companion analysis (4 active bytes, all in lane 7, in words $0, 12, 19, 28$), push 200,000 random message pairs $(x, x \oplus \delta_{\text{in}})$ through r rounds of the compiled permutation, and tally the resulting output differences on a 16-byte (128-bit) projection. A surviving cluster would appear as some projected output difference recurring far more often than chance; at this scale (2×10^5 trials over 2^{128} cells) the chance baseline is essentially zero, so any projected difference recurring even twice is candidate signal (detection floor $\approx 2^{-17}$).

No differential cluster with probability above the $\approx 2^{-17}$ detection floor was found for the minimal-trail input difference at any round from one to eight. This is direct empirical evidence against a fat differential cluster on the trail the MILP bound rides; it is sampling-bounded and does not exclude a fainter cluster (the proven single-characteristic bound is far below any reachable sampling sensitivity), so it complements rather than replaces the analytical single-trail bound.

2.12 RX-difference (rotational-XOR) bias

The plain rotational test above rules out exact rotational symmetry; a finer probe is whether a *rotational-XOR* (RX) difference — a fixed difference imposed between a state and a rotated copy — propagates with detectable bias. We measured, over 500,000 trials per configuration at one

Round	distinct diffs	max recurrence	
1	200,000	1	flat
2	200,000	1	flat
3	200,000	1	flat
4	200,000	1	flat
5	200,000	1	flat
6	200,000	1	flat
7	200,000	1	flat
8	200,000	1	flat

Table 8: Differential clustering on the minimal-trail input difference, 200,000 message pairs per round, 128-bit output-difference projection. Every projected output difference was unique at every round (maximum recurrence 1 against a chance threshold of 2); no differential cluster was detected. Detection floor $\approx 2^{-17}$.

round, the maximum per-output-bit bias across all 2048 output bits (the 5σ detection threshold at this sample size is 0.003536). A single-word sweep over rotation constants $\{7, 13, 17, 19, 31, 37\}$ and injection words $\{0, 8, 16\}$ left 17 of 18 configurations clearly below threshold (range 0.002228–0.002796); the lone marginal exceedance (`rx_rot=17`, `inj_word=8`: bias 0.003572, one bit of 2048, $\approx$1.01× threshold) is consistent with the expected false-positive rate. A heavy-mode sweep over all 63 non-trivial rotation constants for four multi-word injection blocks (252 configurations) failed to reproduce it: every configuration stayed below threshold, the highest bias observed being 0.003472 (1.8% below the 5σ threshold). No RX-difference distinguisher was detected.

2.13 Impossible differentials

For each of four input-bit positions $(0, 64, 512, 1984)$ and four output-byte positions $(0, 31, 127, 255)$, 200,000 random pairs $(x, x \oplus \delta)$ were pushed through the reduced-round permutation and the output-difference histogram at the chosen byte was tallied; any non-trivial output difference with count zero is an impossible-differential candidate. At both one and two rounds every completed configuration was clean (0/255 non-trivial differences missing). No impossible-differential candidate was found.

2.14 Boomerang / BCT

The full 256×256 Boomerang Connectivity Table of the S-box was computed by enumeration. Its maximum entry (excluding the trivial $(a, 0)$ and $(0, b)$ rows) is 6, at $(a, b) = (\texttt{0x01}, \texttt{0x05})$, with non-zero entries distributed as 6 at 510 pairs, 4 at 255 pairs, 2 at 31,620 pairs. A BCT maximum of 6 exceeds the DDT-flat ideal of 2, giving a per-S-box boomerang switch probability of $2^{-5.42}$ — a property of the S-box alone and round-independent. An empirical 50,000-trial boomerang quartet experiment with input difference $\alpha = \texttt{0x01}$ at byte 0 gave a maximum output-difference count of 232 at one round (uniform expectation 195.3, ratio 1.19) and 229 at two rounds (ratio 1.17), both well below the 2.0 anomaly threshold. The elevated S-box BCT entry does not propagate to a full-permutation boomerang distinguisher at any tested round count.

2.15 Differential-linear

This test splits the permutation into a top and bottom half and so requires at least two rounds. At two rounds, five (r_0, δ, γ) configurations were probed over 200,000 trials each (the 5σ bias threshold at this sample size is 0.011180). The highest observed correlation was |corr| = 0.003380 — less than a third of the threshold — and all five configurations were below it. No differential-linear bias was detected.

2.16 Invariant subspace / S-box-coset structure

An exhaustive enumeration of all (a, g) pairs satisfying $S(a) \oplus S(a \oplus g) = g$ found 282 dimension-one coset hits across 141 distinct generators (of 255) — the candidate seeds of an invariant-subspace trail. None of the 282 survived a full round in either the constant-base probe or a random-base empirical probe (2,000 trials per sampled pair), at one or two rounds. An exhaustive check of all 32,385 dimension-two subspaces found no coset hit at all. The dimension-one hit count is unremarkable for an 8-bit S-box; the decisive observation is that the structure does not survive the round function, so it cannot seed an invariant-subspace trail.

2.17 Structural sanity checks

Independently of the above, the compiled permutation passed a battery of gross structural checks (sampled, not proofs): bijectivity on a sample (2000/2000 distinct outputs), absence of fixed points (0 among 503 states, including the all-zero and all-one states), and a sampled subspace-trail collapse test (random low-dimensional affine subspaces show no rank collapse under the permutation). All passed. These rule out several gross structural weaknesses but do not substitute for an analytical invariant-subspace / subspace-trail study.

3 Scope and honest limitations

The Section 2 foundations are exact proofs. The Section 3 battery is *empirical*: each test rules out a detectable distinguisher of its kind at the sampled scale, and passing is necessary but not sufficient for security. Specifically:

- The rotational test is an empirical distinguisher, not a proof of rotational-XOR security; an RX-trail search with explicit carry probabilities (MILP) is the stronger form and remains future work.
- The integral test in this section is the classical (byte-saturation) form. Its analytical counterpart, the bit-based division-property integral (two-subset CBDP, MILP), is now implemented (Section 2.10): the harness is operational and sound, the round-1 program solves to optimality, and the exhaustive multi-round sweep that would fix the integral-distinguisher length is an ongoing computation rather than future work.
- The cube and zero-sum tests use fixed cube/subspace dimensions; they are not exhaustive ANF-degree bounds.
- The algebraic-degree test samples six variable placements; an attacker chooses the placement freely, so a carefully-positioned higher-order distinguisher is not excluded. The exhaustive form is a division-property degree bound (MILP) over all variable choices, which remains future work and is the highest-value remaining self-conducted analysis.
- The impossible-differential, boomerang, differential-linear, and invariant-subspace probes were run only at one and two rounds (the regime where a structural distinguisher, if present, is cheapest to detect); they bound reduced-round behaviour, not the full eight rounds, and

the impossible-differential runs were additionally truncated by their time limit before every input-bit position completed.

- Algebraic (Gröbner-basis) attacks, the interaction of the permutation with the sponge construction, and—most importantly—adversarial review by independent cryptanalysts are not addressed here.

Taken together with the proven active-S-box bounds in the companion analysis, these results constitute a thorough self-conducted first-pass evaluation: the foundations are proven, and no distinguisher of the tested families survives the full permutation. They are offered as evidence supporting the design, not as a substitute for external cryptanalysis.